ACTIVITY SHEET ANSWERS

Chapter 2 Suggested answers

Prior knowledge

1 True

2 True

3 False: Pure substances consist of one type of particle only.

4 True

5 False: Solids, liquids and gases can be mixtures.

6 True

7 False: When making a solution, the solid is called the solute and the liquid, the solvent.

8 True

9 True

10 True

11 True

12 False: Both elements and compounds are classified as pure substances.

13 True

14 False: Physical techniques are used to separate the components of mixtures.

15 False: Filtration involves separating an undissolved substance from a liquid.

16 True

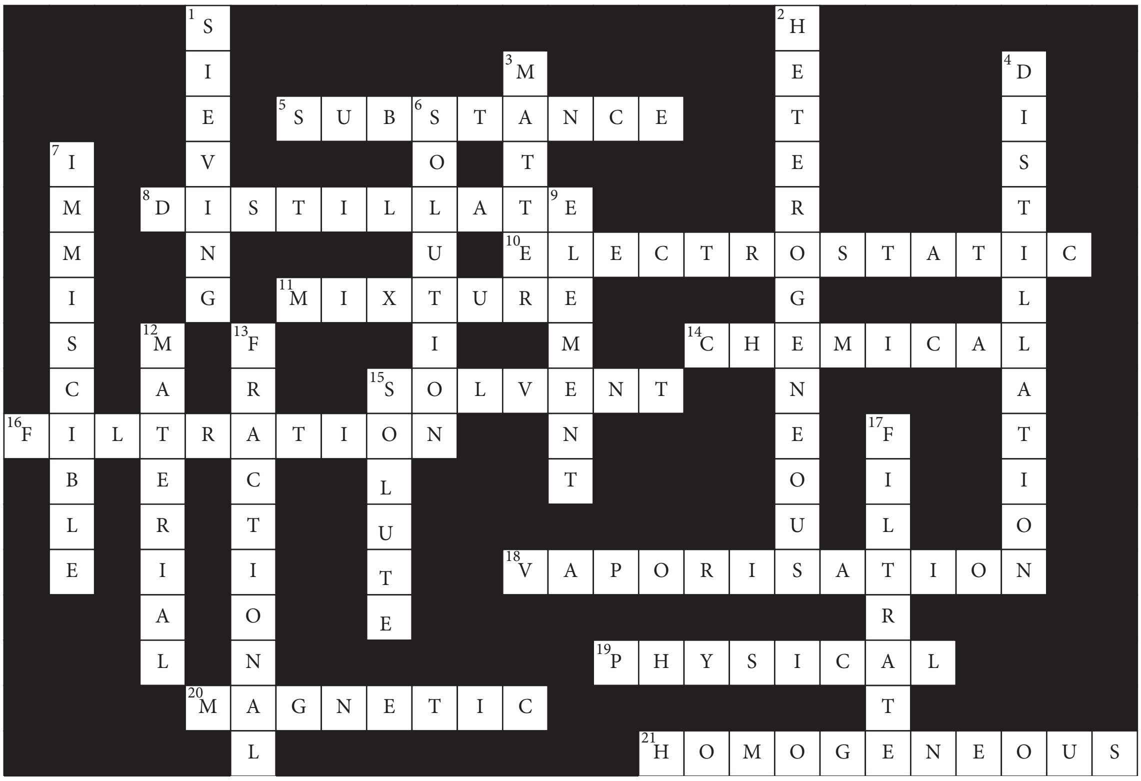
17 True

18 True

19 True

20 False: Distillation involves separating liquids that have different boiling points.

Activity sheet 2.1



Activity sheet 2.4

1 Allow the mixture to settle in a separating funnel. The lighter oil will form a layer on top, which can then be separated by pouring off the mixture below.

2 Heat the remaining mixture in a distillation apparatus to about 80°C. The alcohol will evaporate and can be collected as it condenses back into a liquid.

3 Heat the remaining mixture in the distillation apparatus to about 100°C. The water will evaporate and can be collected as it condenses back into a liquid.

4 After all of the water has evaporated, collect the solid mixture of salt and sugar in the bottom of the flask. Dissolve the mixture in some of the alcohol, and filter out the salt which does not dissolve in alcohol.

5 Heat the remaining solution to about 80°C in the distillation apparatus. The vaporised alcohol can be collected as it condenses back into a liquid, leaving the solid sugar in the bottom of the flask.

Activity sheet 2.5

Physical property	Description of physical property	Separation technique(s) utilising that property
Ductility	ability of the substance to be stretched into a wire	*None*
Solubility	how readily the substance will dissolve in a given liquid	*Filtration (when insoluble)*
Density	the mass of a given volume of the substance	*Separating funnel for two liquids; centrifuging*
Hardness	how resistant the solid substance is to being scratched by other solids	*None*
Electrical conductivity	how readily the substance will conduct electricity	*None*
Electrical charge	the electrical charge on particles of the substance	*Electrostatic attraction*
Thermal conductivity	how readily the substance will conduct heat	*None*
Magnetic effect	attraction of a substance within a magnetic field	*Magnetic separation*
Melting point	distinct temperature at which the solid substance turns into a liquid	*Freezing*
Boiling point	distinct temperature at which the liquid substance turns into a gas	*Distillation; evaporation*
State	either a solid, liquid or gas at the room temperature of 20°C	*Filtration*
Colour	results from particular wavelengths of white light being reflected or transmitted by the substance	*None*
Malleability	ability of the substance to be hammered out into a thin sheet	*None*
Smell	molecules escaping from the surface of the substance detected by the nose	*None*
Particle size	size of the individual particles making up the substance (not at the atomic level)	*Sieving; filtration*

Chapter 2 Revision

1 a → B, b → O, c → M, d → Z, e → X, f → Q, g → N, h → J, i → P, j → Y, k → G, l → H, m → Q, n → U, o → L,p → K, q → C, r → D, s → S, t → F, u → V, v → E, w → A, × → T, y → I, z → W

2 **a** *Physical properties* are those that can be determined without changing the chemical composition of a substance. *Chemical properties* are those that relate to reactions which a substance can undergo.

b *Physical changes* are changes in the physical properties of the substance but not to the chemical composition of the substance. *Chemical changes* are changes in the chemical composition of an existing substance to produce new substance.

c An *element* is a pure substance whose particles consist of only one type of atom. A *compound* is a pure substance whose particles are made of different types of atoms bonded together.

d A *homogeneous mixture* has a uniform composition throughout. A *heterogeneous mixture* has a non-uniform composition.

e A *solute* is the substance that is dissolved in a *solvent* to form a solution.

f A *material* is a substance with particular qualities. A *substance* is a homogeneous material made up of the same type of particle.

g A *filtrate* is the liquid that passes through the filter paper during filtration. A *distillate* is the vapour that condenses back into a liquid and which is collected after distillation.

3

Mixture	Technique	Physical property utilised
Oil and water	*Separating funnel*	*Difference in densities of liquids*
Water and alcohol	*Distillation*	*Difference in boiling points of liquids*
Molten ions	*Electrostatic separation*	*Difference in electrical charge of particles*
Sandy water	*Filtration*	*Difference in state and size of particles*
Crude oil	*Fractional distillation*	*Small difference in boiling points*
Iron sand	*Magnetic separation*	*Differences in responses in a magnetic field*
Boiling rice	*Sieving*	*Difference in particle size*
Salty water	*Vaporisation*	*Much lower boiling point of solvent than solute*

4 strength

density

colour

conductivity

state

solubility

melting point

boiling point

5

Evaporating	Distilling	Filtering	Attracting magnetically	Decanting	Sieving

6 **a** A liquid with an undissolved solid is poured into the filter paper. The liquid passes through the filter paper and the solid (residue) is trapped by the filter paper. The liquid or solution that passes through the filter paper is called the filtrate.

b A solution consisting of two different liquids is heated. The liquid with the lowest boiling point vaporises first and as its vapour passes along the condenser tube it cools, condensing back into a liquid. This is then collected as the distillate.

c Immiscible liquids are not soluble in each other. A mixture of two such liquids separates into two layers if left to stand. The denser liquid will settle on the bottom. When the stopcock at the base of the funnel is opened, the denser liquid runs into the beaker leaving the lighter liquid in the funnel.

d Vaporisation is used to retrieve a solid that has been dissolved in a liquid. The liquid part of the solution (the solvent) is converted into vapour by either boiling the solution or allowing it to evaporate, leaving the solid (the solute) behind.

e Fractional distillation is used to separate liquids that have boiling points that are close together. The mixture is heated and components (fractions) with different boiling points rise up the fractionating column to different heights. The component with the lowest boiling point is collected the top of the column, while the component with the highest boiling point is collected at the bottom of the column.

7 **a** Homogeneous and heterogeneous materials

b Pure substances and homogeneous mixtures (solutions)

c Elements and compounds